

BinderDex

Comprehensive Project Plan

Pokémon TCG Collection Management & Binder Building Platform

Version 3.0 - Final
December 2025

Status: AWAITING APPROVAL

Executive Summary

BinderDex is a Pokémon TCG collection management web application with a unique value proposition: **ChromaDex**, an AI-powered binder page generator that creates aesthetically cohesive layouts based on color analysis and thematic tagging.

Project Overview

Attribute	Details
Target Launch	MVP with core features (Browse, Master Set Tracker, Binder Builder)
Killer Feature	ChromaDex - AI-powered binder page generation (phased rollout)
Monetization	Freemium model with Pro subscription (\$4.99/mo or \$39.99/yr)
Data Source	PokemonTCG/pokemon-tcg-data GitHub repository
Initial Sets	Mega Evolution series (Base Set, Phantasmal Flames)
Platform	Web (desktop-first, mobile responsive)
Backend	Supabase (PostgreSQL + Auth + Storage)

Technical Architecture

Tech Stack

Layer	Technology	Rationale
Framework	Next.js 14 (App Router)	SSR, API routes, Vercel optimization
Language	TypeScript	Type safety, better tooling
Styling	Tailwind CSS + Magic UI	Modern, animated components
State	Zustand	Lightweight, TypeScript-friendly
Database	Supabase PostgreSQL	Managed, generous free tier
Auth	Supabase Auth	Built-in, supports guest → account upgrade
Storage	Supabase Storage	Card images, exports
Hosting	Vercel	Optimal for Next.js
Payments	Stripe	Industry standard
AI	Claude API	Card tagging for ChromaDex

Supabase Free Tier Limits

Resource	Free Tier Limit	Expected Usage
Database	500 MB	~100-200 MB (cards + users)
Auth Users	50,000 MAU	Plenty for launch
Storage	1 GB	~500 MB card images
Edge Functions	500K invocations	ChromaDex generations
Bandwidth	5 GB	Should be fine initially

Data Source Analysis

GitHub Repository

Source: <https://github.com/PokemonTCG/pokemon-tcg-data>

The repository contains JSON files for all card and set data. Each set has its own JSON file in the cards/en/ directory.

Card Data Structure (V2 API Format)

- **id:** Unique identifier (e.g., 'sv1-1', 'mee1-25')
- **name:** Card name
- **number:** Card number in set
- **rarity:** Rarity string (Common, Rare, Illustration Rare, etc.)
- **types:** Array of Pokémon types (Fire, Water, etc.)
- **supertype:** Pokémon, Trainer, or Energy
- **subtypes:** Array (Stage 2, ex, VMAX, etc.)
- **nationalPokedexNumbers:** Array of Pokédex numbers
- **images:** Object with small and large image URLs
- **artist:** Card artist name

Premium Card Identification Strategy

Era	Premium Card Identification
Mega Evolution & Scarlet/Violet	Filter by rarity: 'Illustration Rare', 'Special Illustration Rare'
Pre-Scarlet/Violet	'Rare Ultra' requires manual sorting/tagging (mixed premium content)

Initial Target Sets

The Mega Evolution series is a NEW 2025 series (not the old XY era). We will begin with:

- **Mega Evolution Base Set** - First set in the series
- **Phantasmal Flames** - Second set, released November 14, 2025

Feature Specifications

Feature 1: Card Database Browser

Key Clarification: Browse Cards displays **unique cards only** - no variants. This keeps card counts accurate and UI clean. Variants are only relevant in Master Set Tracker.

Filter System

- **Text search:** Card name, artist
- **Set filters:** Set IDs, series, era
- **Pokémon filters:** Types, supertypes, subtypes, generation (1-9)
- **Rarity filters:** All rarities, isPremium flag
- **Special filters:** isLegendary, isMythical, isPromo

Note: These SAME filters are reused in Binder Builder card picker and ChromaDex generation constraints.

Feature 2: Master Set Tracker

Key Clarification: The Master Set Tracker serves as a **visual guide** showing what the user's physical binder should look like when completed. Users can look at the virtual binder and see cards they are missing in each specific slot.

Binder Layout System

- **Slot configurations:** 9-slot (3×3), 12-slot (3×4), 16-slot (4×4)
- **Dynamic preferences:** Toggle promos, toggle reverse holos
- **Visual states:** Owned (card image), Missing (greyed placeholder with card info)
- **Progress tracking:** Completion percentage, missing cards list

Variant Display

Card variants (Normal, Reverse Holo, First Edition, etc.) are **ONLY displayed in Master Set Tracker**, not in Browse Cards. The layout dynamically generates based on user preferences for that particular master set.

Feature 3: Binder Builder

Key Feature: Split-panel layout with Card Picker on left and Binder Canvas on right. Users can drag cards from the filtered list directly into binder slots.

Card Picker Panel (Left)

- Search bar with autocomplete
- Full filter system (same as Browse Cards)
- Scrollable card grid with drag-and-drop enabled
- Card count indicator

Binder Canvas (Right)

- Visual binder page with droppable slots
- Page navigation controls
- Add/remove page functionality
- Binder settings (name, slot config)

Feature 4: ChromaDex (Flagship Feature)

Key Update: ChromaDex now supports the **same filter system as Browse Cards**, allowing users to constrain generations to specific sets, types, eras, etc.

Generation Modes

Mode	Description
Color	Match cards by dominant color, palette, or color harmony (complementary, analogous, triadic)
Theme	Match cards by AI-generated tags (sunset, forest, battle, peaceful, etc.)
Hybrid	Combine color matching with thematic constraints

Filter-Based Generation

Users can apply Browse Cards filters to constrain which cards ChromaDex can select from:

- *Example:* "Blue page using Water types from Gen 1, Premium cards only"
- *Example:* "Pink Fairy legends from Mega Evolution sets"
- *Example:* "Sunset forest theme from Scarlet & Violet era"

Technical Pipeline

- **Color Extraction:** ColorThief/Vibrant.js for dominant colors & palette
- **AI Tagging:** Claude Vision API for theme/mood/setting tags
- **Scoring Algorithm:** Color distance (CIEDE2000) + tag matching + diversity bonus
- **Arrangement:** Visual flow optimization for binder page layout

Feature 5: Export & Import

Basic Export (Free)

- CSV/JSON format
- Card name, set, number, rarity

Premium PDF Export (Pro)

- Visual binder pages with card images
- Customizable layout and cover page
- Print-ready quality

Import Sources

- TCGPlayer collection export
- Collectr app export
- Custom CSV/JSON upload

Project Phases & Timeline

Phase Overview

Phase	Timeline	Deliverables
Phase 0	Week 1-2	Foundation: Next.js setup, Supabase, data pipeline, CI/CD
Phase 1	Week 3-4	Card Database Browser: Grid view, filters, search, card detail
Phase 2	Week 5-7	Master Set Tracker: Visual binder, preferences, progress tracking
Phase 3	Week 8-9	Binder Builder: Card picker panel, drag-drop, page management
Phase 4	Week 10-13	ChromaDex: Color extraction, AI tagging, generation algorithm, UI
Phase 5	Week 14-15	Export/Import: CSV, PDF generation, TCGPlayer/Collectr parsers
Phase 6	Week 16-17	Community: Binder gallery, sharing, curated binders, polish
Phase 7	Week 18-19	Monetization & Launch: Stripe, feature gating, ads, launch prep

Monetization Model

Tier Comparison

Feature	Guest	Free	Pro (\$4.99/mo)
Browse Cards	✓	✓	✓
Card Search & Filters	✓	✓	✓
Master Set Trackers	1	1	Unlimited
Custom Binders	1 (10 pages)	1 (10 pages)	Unlimited
ChromaDex	✗	3/month	Unlimited
Basic Export (CSV)	✗	✓	✓
Premium Export (PDF)	✗	✗	✓
Import (TCGPlayer, Collectr)	✗	✓	✓
Public Binder Sharing	✗	✓	✓
Ads	✓	✓	✗

Pricing Options

- **Pro Monthly:** \$4.99/month
- **Pro Yearly:** \$39.99/year (33% savings)

Git Workflow & Version Control

Branch Strategy

main ← production deployment

develop ← integration branch

feature/* ← feature development (e.g., feature/card-browser, feature/chromadex)

hotfix/* ← emergency fixes to main

Commit Convention (Conventional Commits)

Type	Description
feat	New feature (e.g., feat(browser): add multi-select type filter)
fix	Bug fix (e.g., fix(master-set): correct reverse holo count)
docs	Documentation changes
style	Formatting (no code change)
refactor	Code restructuring
test	Adding tests
chore	Maintenance tasks

Approval Checklist

Please confirm the following items before implementation begins:

- ☐ **Supabase** as sole backend (Auth + DB + Storage)
- ☐ **No variants in Browse** - only in Master Set Tracker
- ☐ **Binder Builder split layout** with Card Picker panel
- ☐ **ChromaDex filter support** - same filters as Browse
- ☐ **Binder sharing moved to Phase 6** (Community & Polish)
- ☐ **Phase timeline** (~19 weeks) acceptable
- ☐ **Shared filter component** approach (Browse, Builder, ChromaDex)
- ☐ **Initial target sets:** Mega Evolution Base Set & Phantasmal Flames
- ☐ **Premium card strategy:** IR/SIR for modern, manual tagging for older sets
- ☐ **Data source:** PokemonTCG/pokemon-tcg-data GitHub repository

Next Steps After Approval

1. User sets up Supabase project and links MCP
2. Initialize Next.js project with TypeScript, Tailwind, Magic UI
3. Create database schema and run migrations
4. Build data import pipeline for Mega Evolution sets
5. Begin Phase 1: Card Database Browser development

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